Sampling Concepts & Limit Theorems

Population, Sample, Standard Error, and Central Limit Theorem with Practical Examples

Population vs. Sample

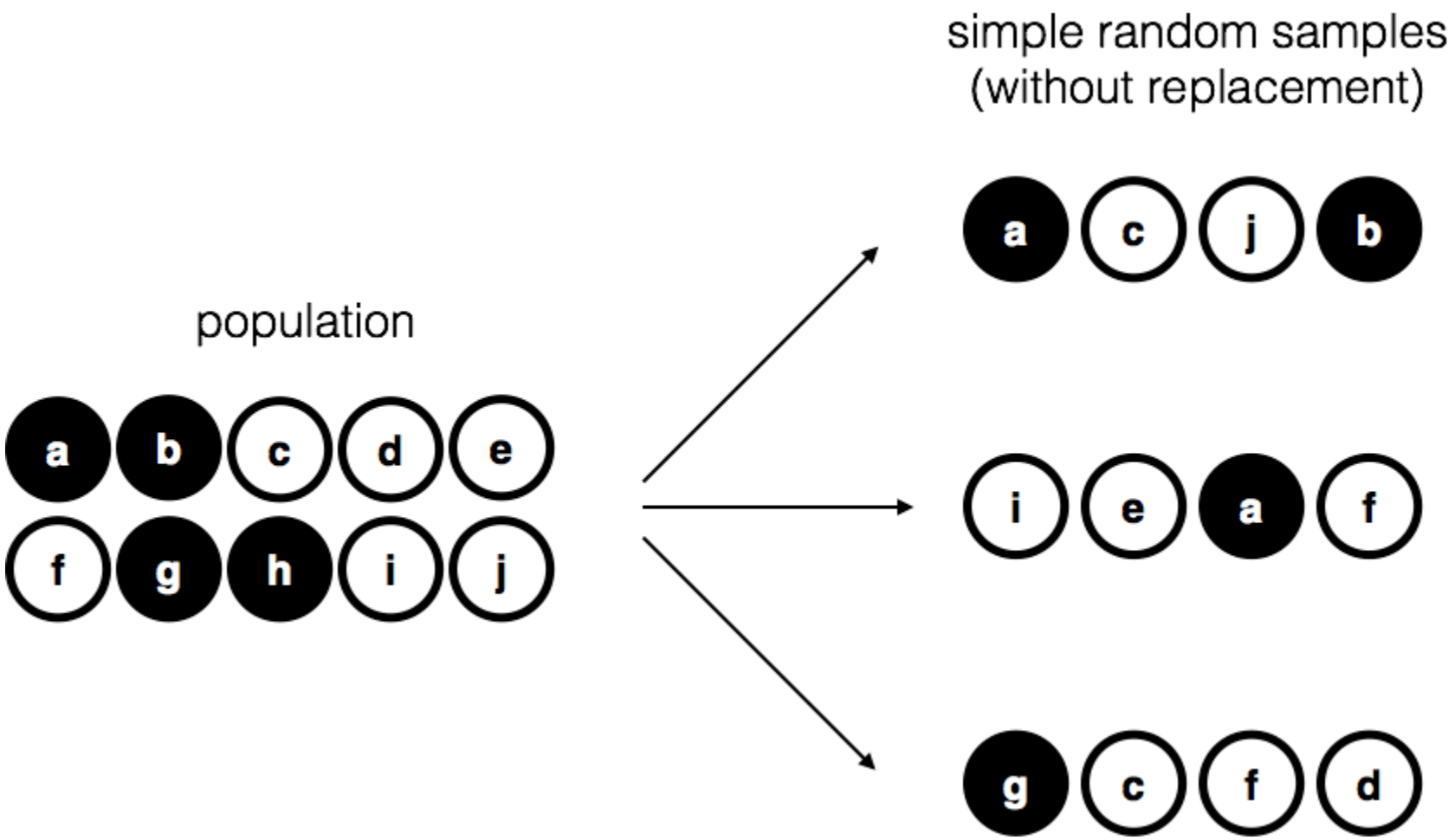
The foundation of statistical inference

Population (N)

- Definition: The entire group of individuals, objects, or measurements that you want to draw conclusions about.
- Characteristics: Parameters (e.g., Population Mean μ, Population Standard Deviation σ).
- Reality: Often too large, costly, or impossible to measure completely.
- Example: All 500,000 lightbulbs produced by a factory in a month.

Sample (n)

- Definition: A specific, smaller subset of the population collected to observe and analyze.
- Characteristics: Statistics (e.g., Sample Mean $\bar{x}$, Sample Standard Deviation s).
- Reality: Must be representative and randomly selected to avoid bias.
- Example: 100 randomly chosen lightbulbs selected for quality testing.


simple random samples
(without replacement)
population
a b c d e
f g h i j
a c j b
i e a f
g c f d

Understanding Standard Error (SE)

Measuring estimate precision

What is Standard Error?

Standard Error measures the variability or dispersion of a sample statistic (usually the sample mean) across multiple hypothetical samples. It quantifies how much your sample estimate is expected to deviate from the true population parameter.

The Formula for Standard Error of the Mean (SEM):

$$SE = \sigma / \sqrt{n}$$

Where: σ = Population Standard Deviation | n = Sample Size

Key Takeaway: As sample size (n) increases, the Standard Error decreases. Larger samples lead to more precise estimates.

Standard Error: Practical Example

Calculating precision step-by-step

Scenario: Lightbulb Lifespan

- A factory produces lightbulbs with a known population standard deviation (σ) of 100 hours.
- A quality control inspector randomly pulls a sample of 25 lightbulbs (n = 25) to test.
- Problem: Find the Standard Error of the Mean (SEM) for this sample size, and see what happens if we increase n to 100.

Calculation Steps:

Case 1 (n = 25): SE = 100 / √25 = 100 / 5 = 20 hours

Case 2 (n = 100): SE = 100 / √100 = 100 / 10 = 10 hours

Interpretation: Quadrupling the sample size from 25 to 100 cut our standard uncertainty exactly in half (from 20 hours down to 10 hours).

Central Limit Theorem

Statistics and Probability – MAS2001

The Central Limit Theorem (CLT)

The bridge to modern data analysis

Core Definition

The Central Limit Theorem states that if you take sufficiently large samples from a population, the distribution of the sample means will follow a normal distribution (bell curve)—regardless of the population's underlying shape (even if it is heavily skewed, uniform, or bimodal).

Three Critical Rules of CLT:

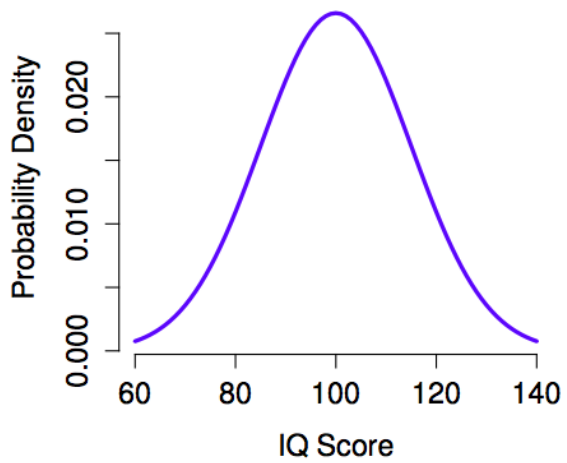
1. Center Rule: The mean of the sample means will equal the population mean ($\mu_{\bar{x}} = \mu$).
2. Spread Rule: The standard deviation of the sample means will equal the Standard Error ($\sigma_{\bar{x}} = \sigma / \sqrt{n}$).
3. Shape Rule: As sample size n increases, the shape becomes normal. If $n \geq 30$, normality is assumed safely.

Note:

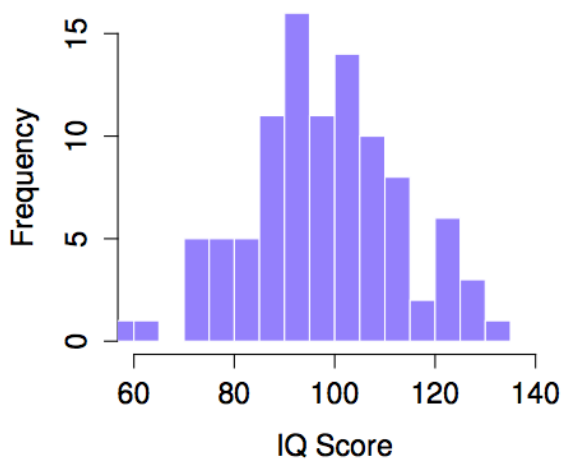
1. For a population with any distribution, if $n > 30$, then the sample means will have a distribution that can be approximated by a normal distribution with mean μ and standard devisation $\frac{\sigma}{\sqrt{n}}$ where n is the sample size.

2. If $n \leq 30$ and the original population has a normal distribution, then the sample means have a normal distribution with mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$,where n is the sample size

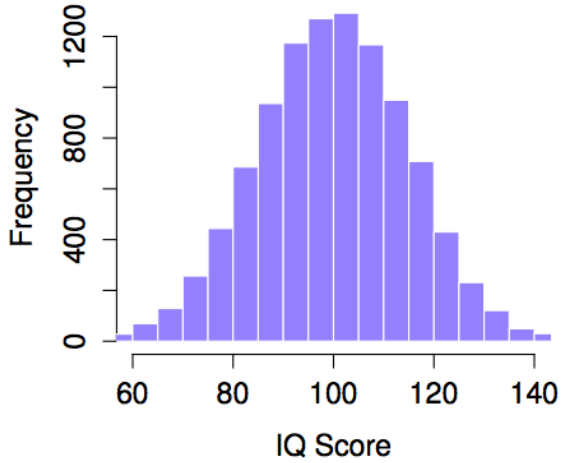
3. if $n \leq 30$ and the original population does not have a normal distribution, then we cannot apply the CLT.


Probability Density
0.000
0.010
0.020
60
80
100
120
140
IQ Score

(a)

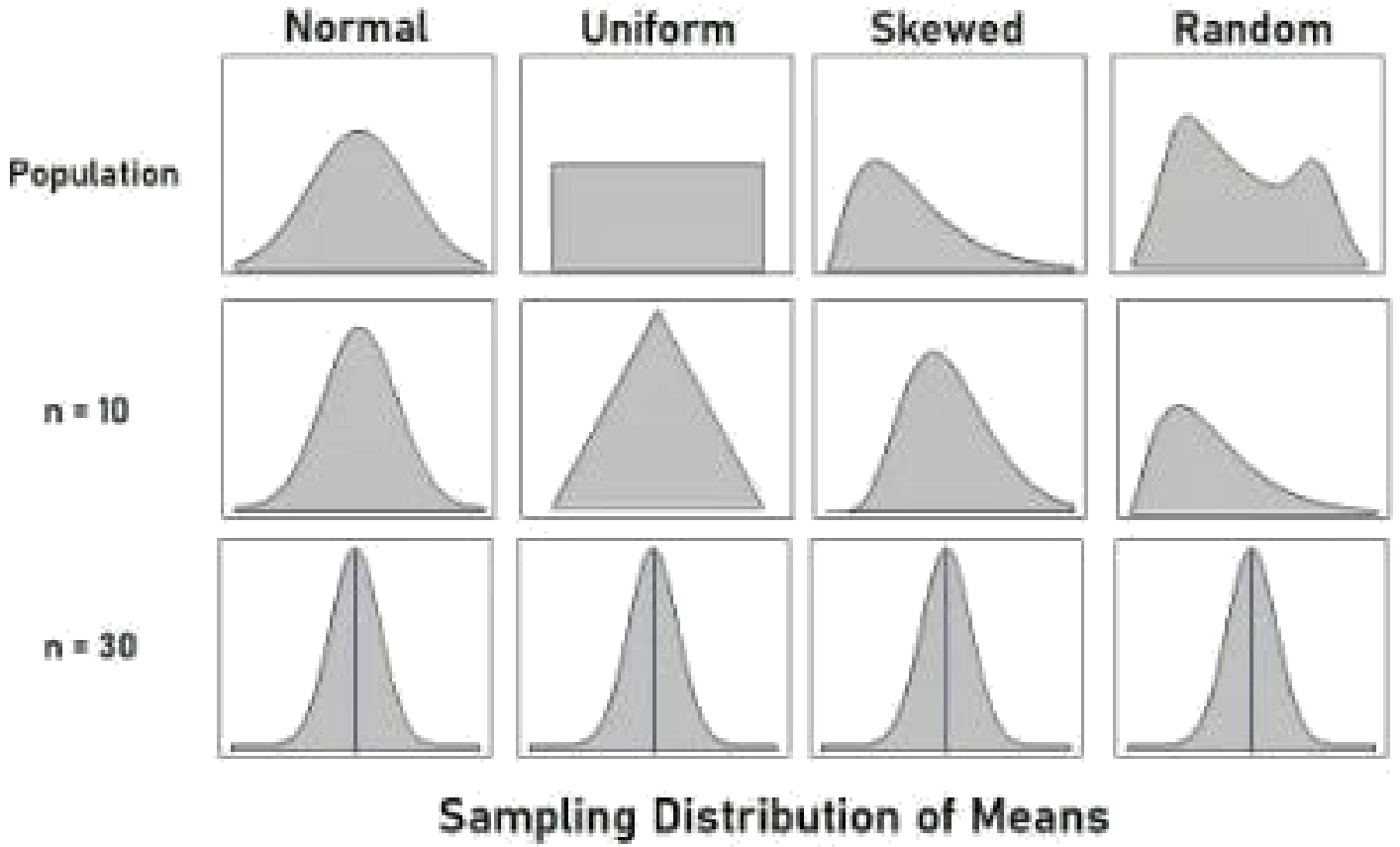

Frequency
0
5
10
15
60
80
100
120
140
IQ Score

(b)


Frequency
0
400
800
1200
60
80
100
120
140
IQ Score

(c)

CENTRAL LIMIT THEOREM



Central Limit Theorem: Practical Example

Applying normality to skewed distributions

Scenario: Bank ATM Wait Times

- The wait times at a bank ATM follow a heavily right-skewed distribution (most people finish fast, a few take a very long time).
- Population parameters: Mean wait time (μ) = 4 minutes, Standard Deviation (σ) = 2 minutes.
- Problem: If we track a random sample of 36 customers (n = 36), what is the probability that their average wait time is greater than 4.5 minutes?

Central Limit Theorem: Practical Example

Solution Using CLT Steps:

1. Apply Normality: Since n = 36 (which is ≥ 30), the sampling distribution of means is Normal, even though individual wait times are skewed.

2. Calculate Standard Error: SE = $2 / \sqrt{36} = 2 / 6 = 0.333$ minutes.

3. Find Z-Score: $Z = (\bar{x} - \mu) / SE = (4.5 - 4) / 0.333 = 0.5 / 0.333 = 1.50$.

4. Determine Probability: Looking up Z = 1.50 on a standard normal table gives an upper-tail probability of approximately 6.68%. Therefore, there is only a 6.68% chance a sample of 36 customers averages over 4.5 minutes.

Example:

The amount of impurity in a batch of a chemical product is a random variable with mean value 4.0 g and standard deviation 1.5 g. If 50 batches are independently prepared, what is the approximated probability that the average amount of impurity in these 50 batches is between 3.5 g and 3.8 g?

Example:

1. The amount of impurity in a batch of a chemical product is a random variable with mean value 4.0 g and standard deviation 1.5 g. If 50 batches are independently prepared, what is the approximated probability that the average amount of impurity in these 50 batches is between 3.5 g and 3.8 g?

Solution:

Given $\mu = 4, \quad \sigma = 1.5, \quad n = 50$

Using CLT, the average will follow the normal distribution with mean (μ)= 4 and SD = $\frac{\sigma}{\sqrt{n}} = \frac{1.5}{\sqrt{\{50\}}}$

Then,

$$P(3.5 < \bar{X} < 3.8) = P\left(\frac{3.5-4}{\frac{1.5}{\sqrt{\{50\}}}} < \frac{\bar{X}-4}{\frac{1.5}{\sqrt{\{50\}}}} < \frac{3.8-4}{\frac{1.5}{\sqrt{\{50\}}}}\right) = P(-2.36 < Z < -0.4)$$

$$\Rightarrow P(3.5 < \bar{X} < 3.8) = P(0 < Z < 2.36) - P(0 < Z < 0.4) = 0.4909 - 0.3264 = 0.1644$$

Example:

2. A factory produces LED bulbs. The lifespan of these bulbs is heavily skewed. The mean life span is 50,000 hours with a standard deviation of 8,000 hours. A quality control inspector tests a random sample of 64 bulbs. What is the probability that the average lifespan of this batch is less than 48,000 hours?

Example:

2. A factory produces LED bulbs. The lifespan of these bulbs is heavily skewed. The mean life span is 50,000 hours with a standard deviation of 8,000 hours. A quality control inspector tests a random sample of 64 bulbs. What is the probability that the average lifespan of this batch is less than 48,000 hours?

Solution:

Given $\mu = 50000, \quad \sigma = 8000, \quad n = 64$

Using CLT, the average will follow the normal distribution with mean (μ)= 50000 and

SD = $\frac{\sigma}{\sqrt{n}} = \frac{8000}{\sqrt{\{64\}}}$

Then,

$$P(\bar{X} < 48000) = P\left(\frac{\bar{X} - 50000}{\frac{8000}{\sqrt{\{64\}}}} < \frac{48000 - 50000}{\frac{8000}{\sqrt{\{64\}}}}\right) = P(Z < -2)$$

$$P(\bar{X} < 48000) = P(-\infty < Z < 0) - P(0 < Z < 2) = 0.5 - 0.4772 = 0.0228$$

Example:

3. The average life of a machine is 7 years with a standard deviation of 1 year. If the lives of these machines follow the normal distribution, find the probability that the average life of 9 random samples of such machines falls between 6.4 years to 7.2 years.

Example:

3. The average life of a machine is 7 years with a standard deviation of 1 year. If the lives of these machines follow the normal distribution, find the probability that the average life of 9 random samples of such machines falls between 6.4 years to 7.2 years.

Solution: Given a population of machines and life time of machines follows a normal distribution with average life (μ) = 7 and SD (σ) = 1.
Sample size (n) = 9
Using CLT we say that average life of the samples also follows the normal distribution with mean (μ)= 7 and SD = $\frac{\sigma}{\sqrt{n}} = \frac{1}{3}$.
Then,

$$P(6.4 < \bar{X} < 7.2) = P\left(\frac{6.4-7}{\frac{1}{3}} < \frac{\bar{X}-7}{\frac{1}{3}} < \frac{7.2-7}{\frac{1}{3}}\right) = P(-1.8 < Z < 0.6)$$

$$P(6.4 < \bar{X} < 7.2) = P(-1.8 < Z < 0) + P(0 < Z < 0.6)$$

$$P(6.4 < \bar{X} < 7.2) = 0.4641 + 0.2257 = 0.6898$$

References:

1. Devore, J. L. (2016). *Probability and statistics for engineering and the sciences* (9th ed.). Cengage Learning.
2. Gupta, S. C. & Kapoor, V. K. (2002). *Fundamentals of Mathematical Statistics* (10th ed.). Sultan Chand & Sons.